

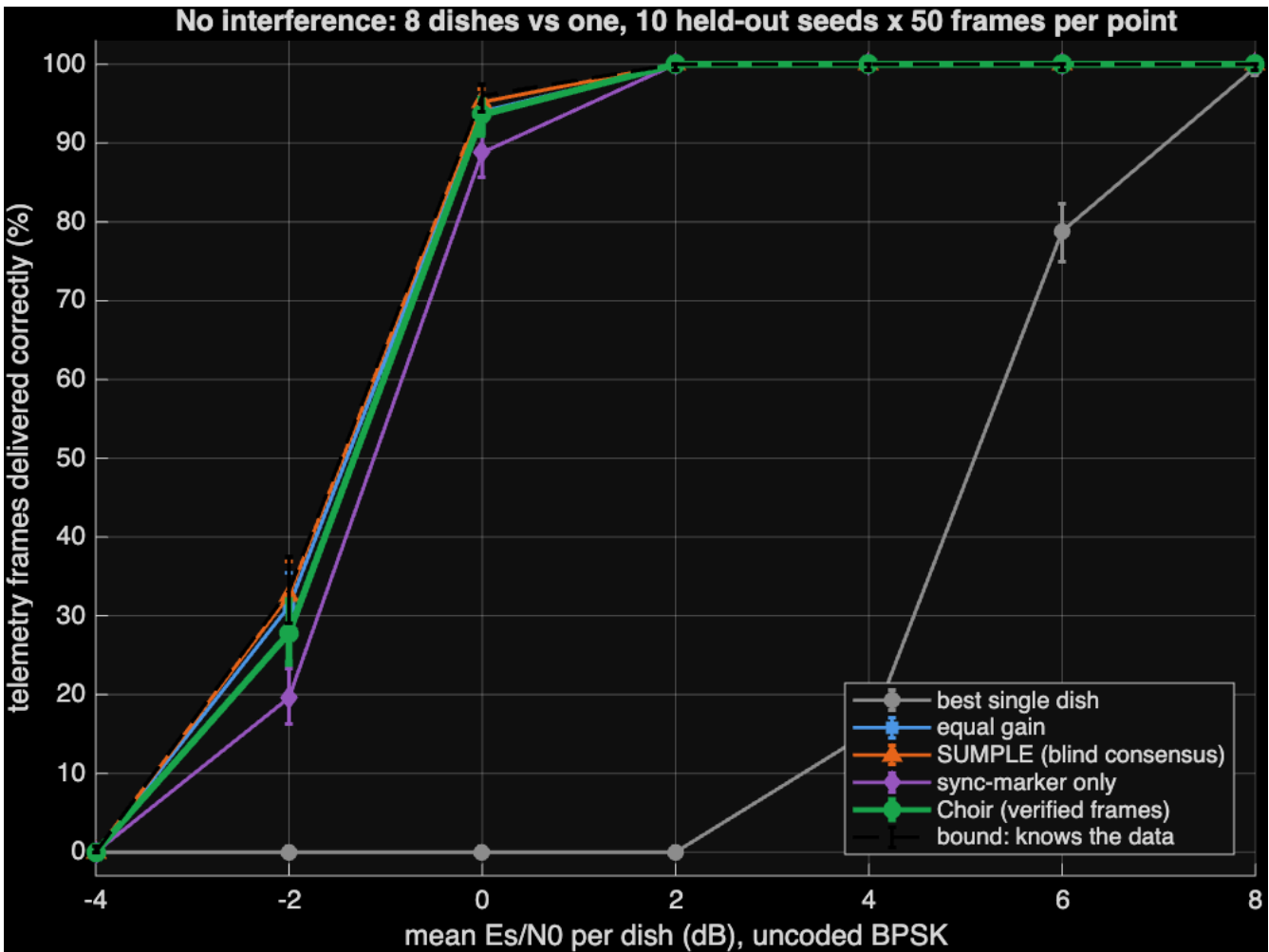
Choir delivered 88.5% of telemetry frames when a look-alike spacecraft was 10 dB stronger

Choir is an autonomous MATLAB receiver for 8 simulated dishes. It calculates the dish weights only from frames that pass three checks. The telemetry is real NASA Curiosity (MSL) data; the radio link is simulated. With the same samples, JPL's SUMPLE method delivered 0%.

The project has two parts: a MATLAB receiver with 8 simulated dishes and a breadboard demo with 3 LEDs

Part	Content	Evidence
Track 1 task	Find, clean, and decode weak radio signals with no human tuning	Required: filter, acquire or re-acquire, decoded telemetry, before/after spectra
Software: Choir MATLAB R2026b	Real NASA Curiosity (MSL) telemetry in CCSDS frames goes to 8 simulated dishes with hidden faults. The dish weights come only from frames with a correct CRC, the correct spacecraft ID, and the next frame count The telemetry is real. The radio link is simulated. No benchmark radio clips are used	Measured results on slides 3 to 9
Hardware: Choir on a breadboard Raspberry Pi Pico	3 LEDs send the same Morse frame (sync, ID C, counter, MATLAB, checksum) in sequence to one photoresistor. In each round, a hidden fault adds flicker to one LED, changes its payload, or sends a look-alike frame on 2 of 3 LEDs	Live: raw and filtered light signal, majority vote and Choir checks, decoded message in the terminal
Choir swarm add-on	Several receivers find the carrier frequency, phase, timing, and gain together (particle swarm). They use a known training message	Slide 12

One dish needs approximately 8 dB to decode a frame; 8 dishes decoded all frames at 2 dB each



99.6%
best single dish at 8 dB

15.2%
best single dish at 4 dB

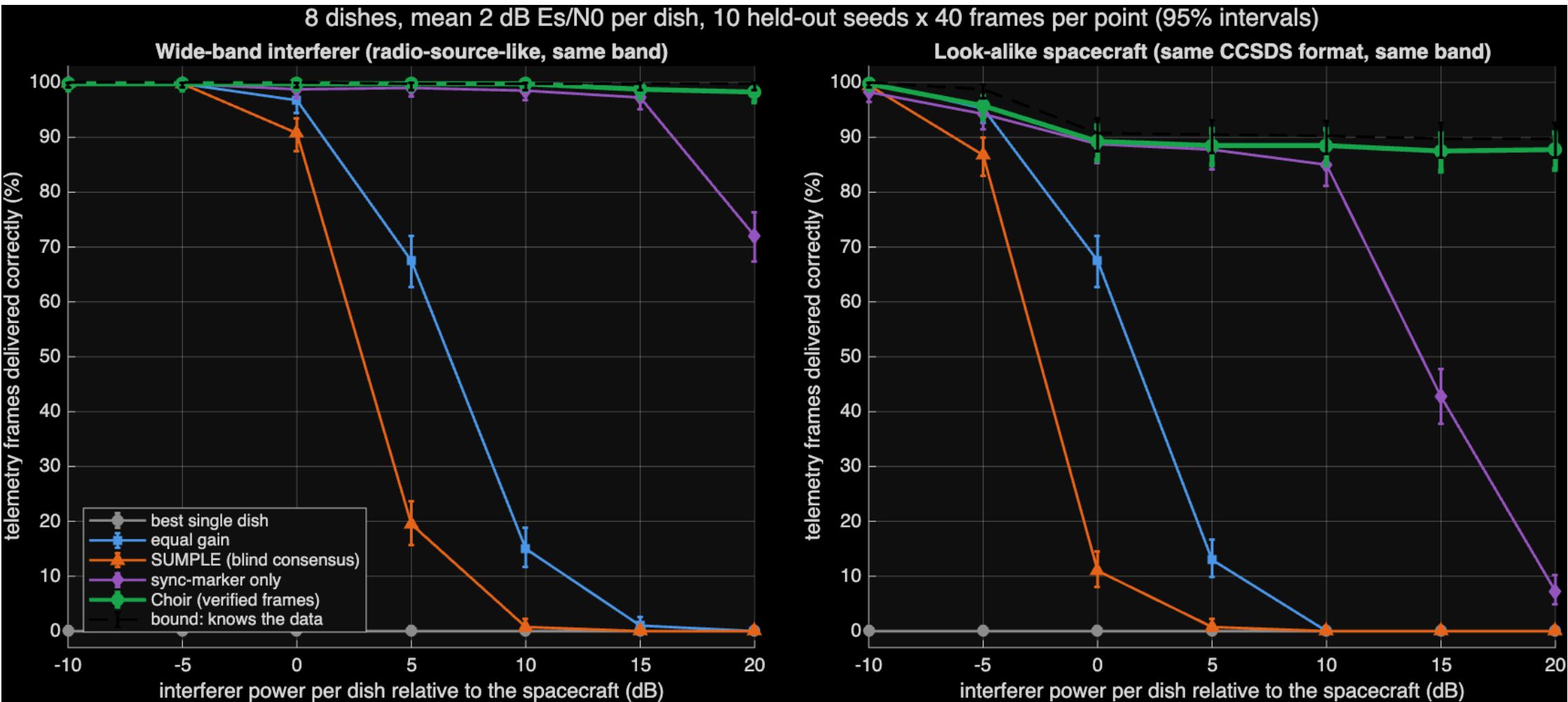
100%
Choir with 8 dishes at 2 dB

Gray line: the best single dish. Without interference, the other 8-dish receivers (equal gain, SUMPLE, sync-marker only) give almost the same results as Choir.

Choir operates in four states without operator input and calculates the dish weights only from verified frames

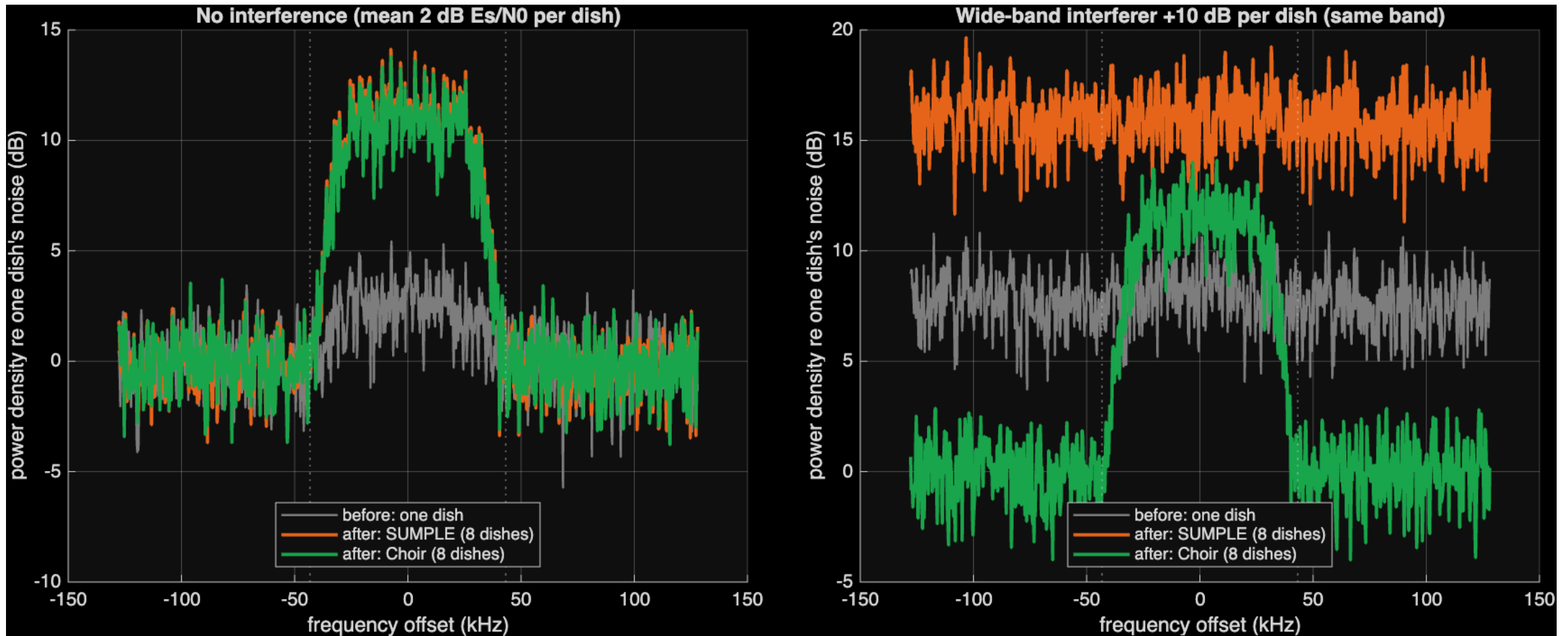
State	Function	Next state
SEARCH	Squares the BPSK signal to show the carrier as a tone. Adds the spectra of all 8 dishes	ALIGN, when a spectral peak is 12 times the median
ALIGN	Finds the sync marker in one frame period on each dish. This gives the frame timing and the delay of each dish Rejects frames with a different spacecraft ID	DECODE, when a frame passes the three checks
DECODE	Uses the weights $w = Q^{-1}h$. h comes only from frames with a correct CRC, spacecraft ID 42, and the next frame count. Q is the rest of the received signal A dish that does not match the verified frames is re-timed or removed	FALLBACK, after 3 bad frames in sequence
FALLBACK	Re-times each dish on the sync marker. Uses equal weights	DECODE, when a frame passes the checks. SEARCH, after 8 bad frames in sequence

With a look-alike spacecraft 10 dB stronger, Choir delivered 88.5% of frames and SUMPLE delivered 0%

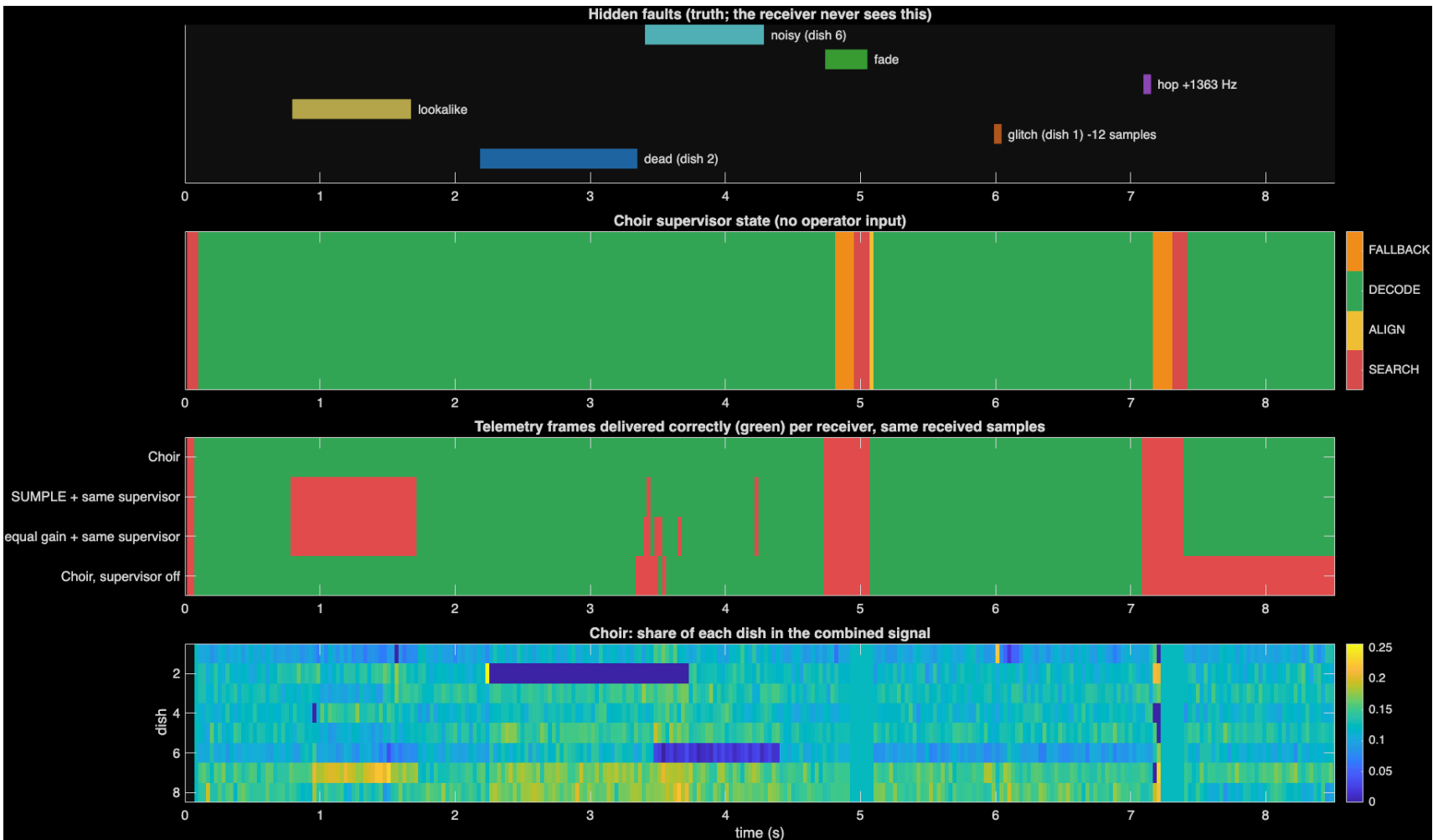


Right panel: look-alike (same CCSDS format and band, ID 21). Sync-marker only: 85.0% at +10 dB, 7.3% at +20 dB (Choir 87.8%). Frames logged with incorrect data: 0. 10 new seeds x 40 frames per point. Data: results/interference_summary.csv

With a +10 dB wide-band source, the Choir output noise floor was 0.1 dB and the SUMPLE output noise floor was 16.0 dB



With six hidden faults per run, Choir delivered 90.6% of frames and recovered 284 ms after a carrier hop



77.7%

SUMPLE with the same supervisor
(mean of 20 runs)

40.0%

Choir with the supervisor off

0

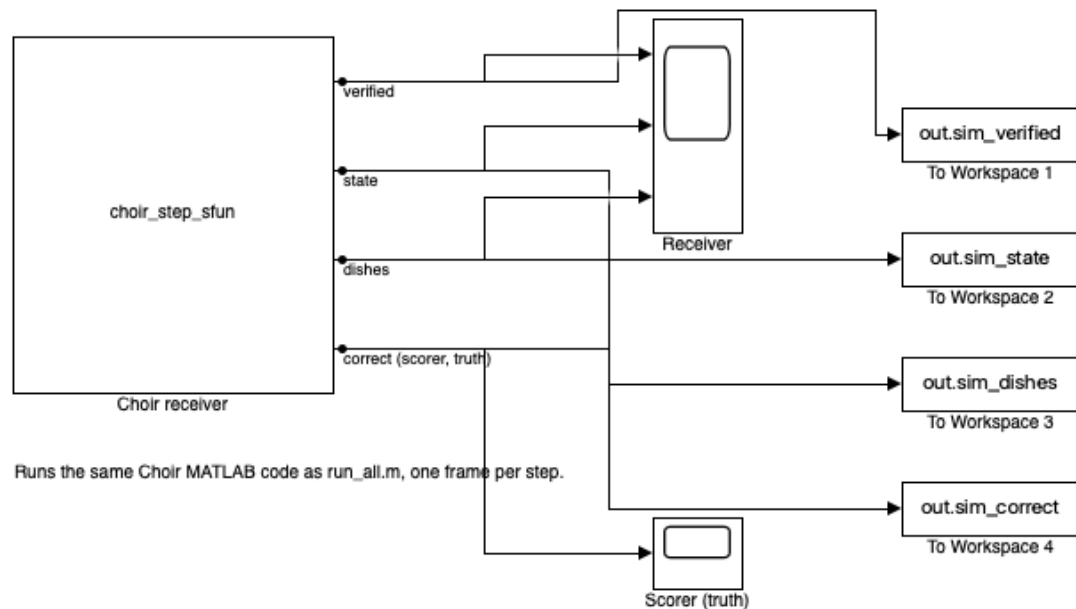
frames logged with incorrect data

Choir recorded each hidden fault in its log within 2 frames (57 ms) and gave the cause

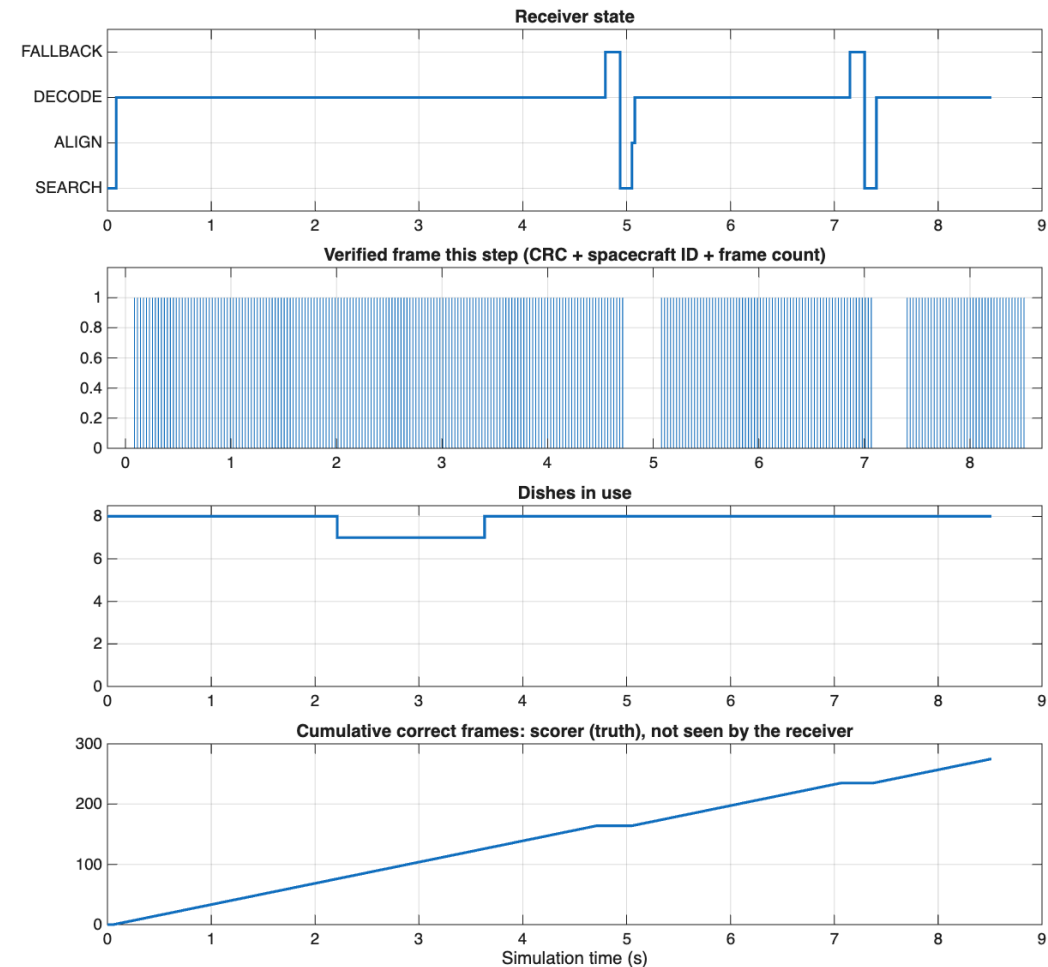
time (s)	state	event	reason (program output)
0.7950	DECODE	interference	a strong extra source reaches 8 dishes (21 dB over noise); weights now cancel it
2.2130	DECODE	dish down	dish 2 stopped matching verified frames (fit 0.80 -> 0.01) and was not found nearby: excluded
3.4050	DECODE	noisy dish	dish 6 is 9 dB noisier than the floor; its weight drops
3.4900	DECODE	noisy dish	dish 6 is 10 dB noisier than the floor; its weight drops
3.6320	DECODE	dish back	probe found dish 2 again (fit 0.83, timing +0 samples): re-included
4.7950	FALLBACK	safe mode	3 bad frames in a row: re-aligned every dish on the sync marker, equal weights
4.9370	FALLBACK	lost	8 bad frames in a row: full carrier re-search
5.0510	SEARCH	carrier found	+64.4 Hz (spectral peak/median 197)
5.9870	DECODE	dish re-aligned	dish 1 stopped matching verified frames (fit 0.75 -> 0.21); timing moved -12 samples, fit now 0.55
7.1510	FALLBACK	safe mode	3 bad frames in a row: re-aligned every dish on the sync marker, equal weights
7.2920	FALLBACK	lost	8 bad frames in a row: full carrier re-search
7.4060	SEARCH	carrier found	+1440.5 Hz (spectral peak/median 1206)

Hidden faults (truth, not given to the receiver): look-alike from 0.79 s, dish 2 fails from 2.19 s, dish 6 noise $\times$ 10 from 3.41 s, fade on all dishes from 4.74 s, clock error on dish 1 at 5.99 s, carrier hop of +1363 Hz at 7.09 s. Each fault has a log entry within 2 frames.

The same receiver code runs in Simulink, one frame per step, and delivered 275 of 300 frames, the same as in MATLAB



choir_sim.slx: Choir receiver in Simulink, one frame per step (seed 301, 300 frames, 2 dB)



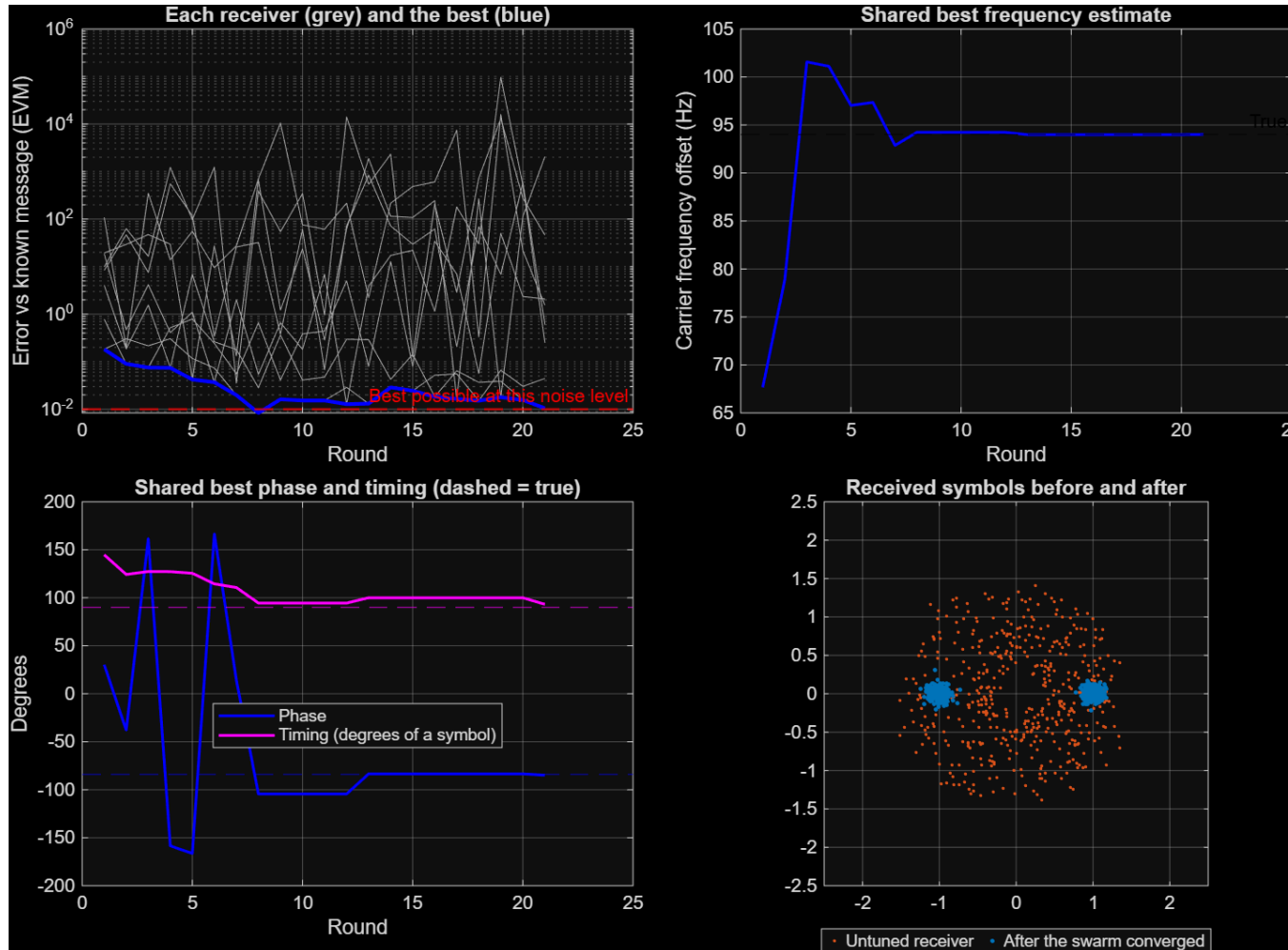
In the breadboard demo, a look-alike frame on 2 of 3 LEDs is designed to win the majority vote; Choir rejects it because of its ID

Hidden fault (one per round)	Signal at the photoresistor	Majority vote uses the most common frame	Choir accepts only verified frames
Flicker on one LED	One copy has errors from flicker	The 2 correct copies agree: correct	The copy with errors fails the checksum; a correct copy is logged
Changed payload	One copy has incorrect letters and the old checksum	The 2 correct copies agree: correct	That copy fails the checksum; a correct copy is logged
Look-alike on 2 of 3 LEDs	Two copies with ID Q, payload DECOY, and a correct checksum	DECOY wins 2 to 1: incorrect	Both copies are rejected (ID Q, not C); the correct copy is logged

No receiver separated the spacecraft from a look-alike with a similar arrival pattern (similarity 0.81)

Limitation	Measured effect	Next step
Look-alike with a similar arrival pattern	Seed 304, similarity 0.81 (other seeds 0.16–0.58): all receivers failed, including the bound that uses the transmitted bits	Separate the signals by frequency or code, not by dish weights
Weak signal without interference	Choir 27.8%, SUMPLE 32.6% at –2 dB per dish	None: the advantage of Choir is resistance to interference and faults
Simulated radio link	The telemetry is real NASA MSL data; dishes, noise, and faults are simulated	Recordings from several real stations
Uncoded BPSK	One dish needs approximately 8 dB; LDPC or turbo codes need several dB less	Add coding; the receiver logic stays the same

The swarm receivers found each carrier offset within 0.6 Hz and decoded two of three test messages with 0 bit errors



39.5 Hz

found for a 40 Hz offset; HELLO FROM MARS, 8 dB: 0 bit errors

99.7 Hz

found for a 100 Hz offset; HI TEAM, 6 dB: 0 bit errors

-179.4 Hz

found for a -180 Hz offset; 4 dB: 3 of 160 bits wrong (BPSK theory: approximately 2)

The figure shows one other run (results/swarm_log.csv).
The three values are the runs in the swarm README.